For Examiner's Use

3 **(a)** A pellet is fired straight through a board that is 10.0 cm thick. It strikes the board with a speed of 120 m s^{-1} and emerges with a speed of 85 m s^{-1}.

Calculate

(i) the average deceleration of the pellet as it passes through the board,

deceleration = ... m s^{-2} [2]

(ii) the time taken for the pellet to pass through the board.

time taken = ... s [1]

(b) After passing through the board, the pellet in **3(a)** enters at a speed of 85 m s^{-1} into a wooden block that is initially at rest on a frictionless table.

The pellet remains in the block after impact.

(i) Explain why the principle of conservation of momentum can be applied in this collision.

.....

..... [1]

For Examiner's Use

(ii) Given that the masses of the pellet and the wooden block are 0.40 g and 50.0 g respectively, determine the speed of the block and pellet after impact.

speed = ... $m\ s^{-1}$ [2]

(iii) Explain, with appropriate calculations, whether the collision is elastic or inelastic.

.....

..... [2]

[Total: 8]